John Ian So

johnianrso@berkeley.edu | johnianrso | johnrso | (925) 336-7553

Skills

Languages:
Java,
C,
C++,
JavaScript,
Python,
Ruby,
RISC-V,
SQL,
HTML/CSS.

Frameworks:
ReactJS,
Flask,
Android Studio,
OpenMP.

Robotics:
ROS,
OpenCV,
Keras,
TensorFlow,
Arduino,
ESP.

Tools:
Git,
Unix Shell,
IntelliJ.

Circuit Design:
Altium Designer,
Multisim,
Autodesk Eagle.

Education

University of California, Berkeley **GPA: 4.00 / 4.00**

B.S. Electrical Engineering and Computer Science, May 2023

Courses Completed: Structure & Interpretation of Computer Programs, Data Structures, Machine Structures, Discrete Mathematics & Probability Theory, Designing Information Devices & Systems I, Multivariable Calculus, Self-Driving Cars

Courses in Progress: Efficient Algorithms & Intractable Problems, Designing Information Devices & Systems II.

Honors: Eta Kappa Nu Mu Chapter (inducted April 2020).

Work Experience

Robotics Intern, PhaseSpace Motion Capture *May 2020-August 2020*

- Researched high contact surface sanitization methods, including UV-C and hypochlorous acid (HOCl).
- Prototyped software, firmware, and hardware using an ESP-32 controller and Firebase for a web interfaced, closed loop control system for a HOCl generator and dispenser system as a response to COVID-19.
- Wrote and submitted a project proposal to the Department of Homeland Security SVIP Program for $200,000 for the HOCl system. Results pending.
- Implemented robot system architecture in ROS Melodic. Modelled a robot frame using URDF and simulated sensors and actuators in Gazebo.
- Designed a navigation software stack based on Real-Time Appearance-Based Mapping, Adaptive Monte Carlo Localization, and visual and IMU odometry.

Lead Intern, PhaseSpace Motion Capture *June 2018-August 2018*

- Led team of 4 in building a robot arm driven by an ESP-32 controller and PhaseSpace's motion capture system.
- Engineered a control system in Arduino to translate 3D vectors to arm joint angles.
- Developed software in Java to translate motion capture data into scaled vector representations and feed the data to the arm's control system.

Activities

Member, Machine Learning at Berkeley *September 2020-Present*

Undergraduate Research Assistant, Biomimetic Millisystems Lab *September 2020-Present*

Course Staff, CS 61A *August 2020-Present*

- Leading tutorials for discussion section four times per week and holding drop-in office hours twice per week to reinforce course concepts, including Python and programming fundamentals.
- Developing software for internal staff tools, including Slack bot integration.

EECS Engineer, Formula Electric at Berkeley *January 2020-Present*

- Engineered PCB designs in Altium Designer for the tractive system indicator and driver interface subsystems of an electric vehicle for the Formula SAE Competition EV division.
- Creating firmware for STM controllers in Mbed OS.

Projects

Mask Vision *August 2020-Present*

- Researched and implemented CNNs using Keras for detection of masks in images.
- Automated preparation of data sets using OpenCV and Keras. Currently looking into transfer learning using the YOLO architecture for real-time recognition of proper and improper usage of masks.

Word Hunt Solver *May 2020*

- Built a model-view-controller program for Word Hunt, a game where players swipe words on a 4x4 board.
- Implemented a solver that finds all possible words on the board within one second using DFS graph traversal.
- Currently deploying as a web application using Heroku, Flask, and ReactJS.

Self-Driving Car Simulation *December 2019*

- Modelled and implemented self-driving car subsystems such as acceleration control, route planning, and steering control in Python using the Duckietown environment, OpenCV, and TensorFlow.
- Created models for linear regression and neural networks. Used PID, A*, and cost maps to plan and run routes.